

contact with the metallic surface of the solid metal part. Any filmy adhering contamination such as oxides or oil on the surface to be soldered will act as barrier to metallic continuity and thus prevent wetting. A drop of solder on a contaminated surface behaves in the same way as an isolated drop of water on a greasy plate.

## Pick and place

Now, paste is applied. The interconnection will fail if the components are not placed properly on the board. Let's see how 'pick and place' takes place. You have to be cautious in this step to avoid drifting. All pick-and-place machines operate in automatic mode. For small-to medium-volume productions, single-arm machines are sufficient for the job. For large-scale production, simultaneous placement machines are required.

The basic principle of 'pick and place' is driven by the movement of vacuum nozzles. Moreover, there is a sensor to recheck picking before proceeding for placement. Precisely, the operation takes place by using a combined setup of hardware and software.

In large-scale automation, the pick-and-place machines are able to handle more than 40,000 components per hour. According to a Frost & Sullivan report, pick-and-place equipment

software account for more than 70 per cent of total SMT software. So try to understand both the hardware and the software aspects of the operation.

## Soldering redefined

Soldering is the most significant step of the process. It marks the difference between SMT and traditional through-hole techniques. Major SMT operations follow either preheating or reflow soldering.

Reflow soldering is the most preferable. Reflow soldering of surface-mount components is advantageous over through-hole methods due to provision of mass soldering, consistency in manufacturing through precise control of process parameters and also flexibility to run small production as well.

You need to know the advantages of this method over the wave soldering. It does not need any extra solder as the solder paste already exists on the board. The paste reflows on heating and solder joints are formed on cooling.

In most cases, the heating is done either using infrared rays or by immersing in the vapour of a liquid boiling at a temperature higher than the melting point of the solder. Usually, liquid fluorocarbons are the liquid of choice.

Other modes of heating in use are hot-air and hot-gas. Try to get some

knowledge about the specific application, advantages and loopholes of each process. For example, application of vapour provides uniform heating but the process may be costly and produce pollutants. As an SMT engineer you will be expected to understand the process optimisation. Keep in mind that reflow soldering often takes more time but is also more versatile from the application point of view.

## Touch-up jobs

Now you are almost at the finishing line of the chip-to-ship cycle. Soldering is done. The raw products are ready. But it is mandatory to clean them and assure their quality before serving to the customers. Cleaning of paste residues is important to avoid corrosion of joints.

Selection of proper cleaning solvent is really a tough job. In most cases, water-based solvents serve well. But for the sake of better understanding, try to get information about other types of solvents also. For example, fluorinated solvents are good cleaner but not used due to environmental hazards.

After cleaning, the finished products are stored and tested. Storage, handling and testing are continued as per the quality standard protocols designed by regulatory bodies like SMEMA.

## Get set go...

Now that you have had a deep understanding of SMT, you are ready to explore the SMT field. So pull up your socks, plan your final-year project in SMT field and find a suitable firm to pursue it. To grab every possible opportunity, surf the websites of related organisations and send requests for your project work. You can even visit the 15<sup>th</sup> Componex Nepcon India 2009, which will be held from 24th to 26th February at Pragati Maidan, Delhi. Perhaps, it will be your first interaction with the industry, but believe me it will be really fruitful. ●

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**A holistic training for entry-level professionals in the field of SMT should cover the basic theory and applications of SMT, process flow, soldering techniques and solder pastes, basic principles of machineries used for SMT, software and hardware related to SMT process, maintenance and quality control procedures and protocols in the SMT process and also storage, shipping and handling of SMT products. ”**

—Avishek Tyagi, regional manager-sales and marketing, American Tec.

